CHAPTER 4 · LESSON 1: WHAT IS AN ALGORITHM? (Pages 43–44)

■ **One-Line Topic Concept:** An Algorithm is a step-by-step set of simple instructions followed in order from START to STOP to solve a problem or complete a task!

ANATOMY OF AN ALGORITHM: FROM START TO STOP

[■ START] ➔ **Step 1:** Clear Action ➔ **Step 2:** Next Action ➔ **Step 3:** Finish Action ➔ **[■ STOP]**
• **Rule 1:** Every algorithm MUST have a clear **START** and **STOP**.
• **Rule 2:** Steps must be followed in **exact sequential order** (1, 2, 3...).
• **Rule 3:** A computer cannot guess; instructions must be simple and precise!

Activity 1 · Complete the Official Definition of Algorithm:

Word Bank: instructions • step-by-step • STOP • START • task

An algorithm is a **step-by-step** set of **instructions** designed to perform a specific **task**. It always begins with **START** and ends with **STOP**.

Activity 2 · Write [T] for True or [F] for False:

[**F**] 1. An algorithm can start without any beginning step. ➔ *F (False — It must begin with START)*
[**T**] 2. Changing the order of steps in an algorithm can cause wrong results. ➔ *T (True)*
[**T**] 3. Computers follow algorithms given by programmers. ➔ *T (True)*
[**T**] 4. Real-life daily routines are also examples of algorithms. ➔ *T (True)*
[**F**] 5. Skipping a step in an algorithm is completely fine. ➔ *F (False — Every step is necessary)*

Activity 3 · Which of these are Real-Life Algorithms? (Write YES or NO):

• 1. Baking a chocolate cake by following a cookbook recipe ➔ **YES**
• 2. Dreaming while sleeping at night without control ➔ **NO**
• 3. Brushing your teeth in the morning step-by-step ➔ **YES**
• 4. Washing clothes using a washing machine cycle ➔ **YES**

Activity 4 · Answer in Complete Sentences (Spacious Writing Lines):

1. What are the two mandatory words that begin and end every algorithm?
➔ ***Every algorithm must begin with START and end with STOP.***

2. Why is it important to follow steps in exact order?
➔ ***Because if we change the order or skip steps, the task will fail and produce wrong results.***

Activity 5 · Match the Algorithm Rule with Its Purpose:

Algorithm Rule	Why it Matters?	Answer Bracket
1. Clear START point	a) Steps must be easy so computers don't fail	**1 ➔ c (Beginning)**
2. Step-by-Step Order	b) Tells when the problem is solved and finished	**2 ➔ d (Exact sequence)**
3. Definite STOP point	c) Shows exactly where the process begins	**3 ➔ b (Finished)**
4. Simple Instructions	d) Guarantees correct result without confusion	**4 ➔ a (Clear steps)**